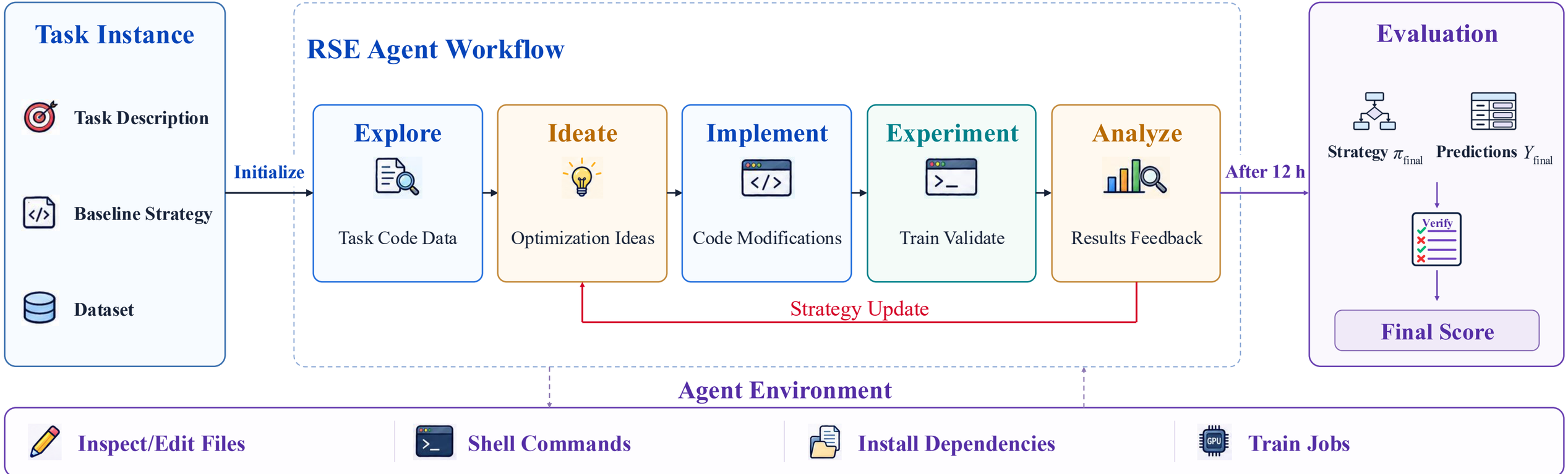

Task Instance
Task Description
Baseline Strategy
Dataset
Initialize
RSE Agent Workflow
Explore
Task Code Data
Ideate
Optimization Ideas
Implement
Code Modifications
Experiment
Train Validate
Analyze
Results Feedback
Strategy Update
After 12 h
Evaluation
Strategy π_{final}
Predictions Y_{final}
Verify
Final Score
Agent Environment
Inspect/Edit Files
Shell Commands
Install Dependencies
GPU
Train Jobs